A Short History of the Printing Press

An original overview written as sample content for the Document Intelligence Assistant. Free to reuse. Dates for early printing are approximate, as is usual for this period.

1. Before Movable Type

For most of history, books were copied by hand. In medieval Europe much of this work was done by scribes in monasteries and, from the 13th century, by professional copyists working for universities and booksellers. A single large book could take months to copy, which made books expensive and rare.

Printing from carved wooden blocks developed much earlier in East Asia. Woodblock printing was used in China by the Tang dynasty, and the **Diamond Sutra** found at Dunhuang, dated to 868 CE, is often cited as the oldest surviving printed book with a date. Woodblock printing worked well for texts that were reprinted many times, but every page required a new carved block.

2. Movable Type in Asia

Around 1040 CE, during the Song dynasty, the Chinese artisan **Bi Sheng** made movable type from baked clay, according to the scholar Shen Kuo, who described the process in his book Dream Pool Essays. Each character was a separate piece that could be arranged, printed, and then reused. Later printers experimented with wooden type, and metal type was also developed.

In Korea, metal movable type was used during the Goryeo dynasty. The **Jikji**, a Buddhist text printed in **1377**, is the oldest surviving book printed with movable metal type. A copy is held by the National Library of France. In the 15th century, King Sejong's court promoted the Korean alphabet, Hangul, which has far fewer characters than written Chinese and is well suited to movable type.

Because Chinese writing uses thousands of characters, movable type did not replace woodblock printing in China for centuries; both methods continued in use.

3. Gutenberg and the European Press

Johannes Gutenberg, a goldsmith from Mainz in Germany, developed a printing system in the 1440s and 1450s that combined several innovations:

- A **hand mould** for casting large numbers of identical metal letters quickly and accurately.
- A metal **alloy** of lead, tin and antimony that melted at a low temperature, cooled quickly and was durable.
- An **oil-based ink** that stuck to metal type better than the water-based inks used for woodblocks.
- A **screw press** adapted from presses used for wine and olive oil, which applied even pressure across the page.

Gutenberg's best-known work is the **Gutenberg Bible**, also called the 42-line Bible because most pages have 42 lines of text in each column. It was completed in Mainz around **1454 to 1455**. About 180 copies are thought to have been printed, some on paper and some on vellum. Fewer than 50 copies survive, many of them incomplete.

Gutenberg borrowed money to fund his work, and a lawsuit brought by his financial backer Johann Fust in 1455 cost him much of his workshop. Fust and Peter Schoeffer, Gutenberg's former assistant, continued printing and in 1457 produced the Mainz Psalter, one of the first books to name its printer and date of printing.

4. The Spread of Printing

Printing spread quickly through Europe, carried by German printers who set up workshops in other cities.

Year (approx.)	Place	Event
1465	Subiaco, Italy	First press in Italy, set up by Sweynheym and Pannartz
1469	Venice	Johann of Speyer granted a privilege to print in Venice
1470	Paris	First press in France, at the Sorbonne
1473 to 1474	Bruges	William Caxton, who learned printing in Cologne, prints the first book in English
1476	Westminster, England	William Caxton sets up the first press in England
1501	Venice	Aldus Manutius publishes small-format books using italic type

Books printed in Europe before 1501 are known as **incunabula** (from the Latin for "cradle"). By 1500, presses were operating in more than 200 European towns and cities and had produced, by common estimates, millions of copies, often given as around twenty million.

Venice became the leading centre of printing by 1500. The Aldine Press of Aldus Manutius published Greek and Latin classics in affordable, portable editions, and its italic typeface allowed more text to fit on each page.

5. Effects on Society

The printing press is often linked to major changes in European society, although historians debate how direct those links were.

- **Religion:** Martin Luther's Ninety-five Theses of 1517 were reprinted rapidly, and printed pamphlets helped spread the ideas of the Protestant Reformation. Luther's German translation of the Bible, with the New Testament published in 1522, reached a wide audience.
- **Science:** printing allowed identical copies of texts, diagrams and tables to circulate, so scholars in different cities could work from the same material and correct errors. Copernicus's On the Revolutions of the Heavenly Spheres (1543) and Vesalius's anatomy text De humani corporis fabrica (1543) depended on careful printed illustrations.
- **Language:** printers favoured particular spellings and dialects, which helped standardize languages such as English and German.
- **Literacy and education:** cheaper books and pamphlets increased the demand for reading skills over the following centuries.

6. Later Developments

The basic hand press changed little for about 350 years. The industrial era then brought rapid change:

- **Around 1800:** iron hand presses, such as the Stanhope press, made printing larger sheets easier.
- **1810s:** Friedrich Koenig developed a steam-powered printing press; The Times of London began printing on a Koenig steam press in 1814.
- **1840s:** Richard March Hoe developed the rotary press, which printed from type on a rotating cylinder and greatly increased speed.
- **1880s:** Ottmar Mergenthaler's Linotype machine cast whole lines of type from a keyboard, transforming newspaper production.
- **20th century:** offset lithography became the dominant commercial printing method, followed by phototypesetting and, from the 1980s, desktop publishing and digital printing.

7. Summary

The printing press did not appear from nowhere: it built on paper-making, woodblock printing and movable type developed in China and Korea, as well as on European metalworking and press technology. Gutenberg's contribution was a practical system for producing type and printed pages in large numbers, which spread across Europe within fifty years and helped reshape religion, science, language and education.